ReentrantLockExample.java:->

```
package com.example.demo;

import java.util.concurrent.locks.Lock;
import java.util.concurrent.locks.ReentrantLock;

public class ReentrantLockExample {
    private final Lock lock=new ReentrantLock();
    private int counter=0;

    public void increment()
    {
        lock.lock();
        try {
            counter++;
            System.out.println(Thread.currentThread().getName() + " counter:
" + counter);
        }finally {
            lock.unlock();

        }
    }
}
```

MainTest.java:->

```
package com.example.demo;

public class MainTest {
    public static void main(String[] args) throws InterruptedException {
        ReentrantLockExample reentrantLock=new ReentrantLockExample();
        Thread t1=new Thread(reentrantLock::increment,"Thread 1");
        Thread t2=new Thread(reentrantLock::increment,"Thread 2");

        t1.start();
        t2.start();

        t1.join();
        t2.join();

    }
}
```

ReadWriteLockExample.java:->

```
package com.example.demo;

import java.util.concurrent.locks.ReentrantReadWriteLock;

public class ReadWriteLockExample {

        private final ReentrantReadWriteLock lock = new
ReentrantReadWriteLock();
        private final ReentrantReadWriteLock.ReadLock readLock =
lock.readLock();
        private final ReentrantReadWriteLock.WriteLock writeLock =
lock.writeLock();
        private int data = 0;

//        write method
        public void writeData( int value)
        {
            writeLock.lock();
            try {
                data = value;
                System.out.println(Thread.currentThread().getName() + "
wrote: " + data);
            } finally {
                writeLock.unlock();
            }
        }

        //read method
        public void readData()
        {
            readLock.lock();
            try{
                System.out.println(Thread.currentThread().getName()+" read:
"+data);
            }finally {
                readLock.unlock();
            }
        }
    }
```

MainTest1.java:->

```
package com.example.demo;

public class MainTest1 {
    public static void main(String[] args) throws InterruptedException {
        ReadWriteLockExample readWriteLock=new ReadWriteLockExample();
        Thread writeThread=new Thread(()->readWriteLock.writeData(49),"Write
Thread");
        Thread reader1=new Thread(readWriteLock::readData,"Reader 1");
        Thread reader2=new Thread(readWriteLock::readData,"Reader 2");

        writeThread.start();
        reader1.start();
        reader2.start();

        writeThread.join();
        reader1.join();
        reader2.join();


    }
}
```